



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

RESIT EXAMINATION JUNE 2023

Course : EC3105 (C Programming)

Time : 2.00 pm – 5.00 pm
(3 hours)

Lecturer : Noor Zuhaili bt Md Yasin

Date: 3 October 2023

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **5** printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Name the stage during which the C compiler processes directives such as #define and #ifdef in the process of building an executable. (1 MARK)

- b. Rewrite the code below using a for loop. (4 MARKS)

```
int i = 1;
while (i <= 10)
{
    printf ("hello\n");
    i=i+2;
}
```

- c. Compute the values of the following C expressions assuming that b, s, e, m, k, z and t are integer variables as declared below. (5 MARKS)

```
int b = 1, s = 3, e = 5, m = 4, k = 10;
int z = 2, t = 5;
```

- i. $z = b + k * z / m \% e - 1$
- ii. $e \% s * m > z \parallel k \% s * b < t$
- iii. $(b < s) \% s + 1$
- iv. $++b \&\& s--$
- v. $m = --b ? -e : ++e$

- d. Evaluate the following expressions, assuming x and y are both declared as integer variables and x = 17 and y = 6. (4 MARKS)

- i. $x / y + 3$
- ii. $x \% y * 4$
- iii. $x - y / 3$
- iv. $x / 5 - y * 2$

- e. List **SIX (6)** properties of command line arguments. (6 MARKS)

- f. List any **FIVE (5)** basic UNIX users commands. (5 MARKS)

2. a. Discuss **THREE (3)** steps for executing the C program. (6 MARKS)

b. Consider the following incomplete C program. Fill in the **SEVEN (7)** blanks according to the descriptions given in the comments in the program. (14 MARKS)

```
int main(void) {
    int a = 5;
    char s[] = "Classical Music";
    char* p = (s + 4);
    char c = 'X';
    double d = 150000;
    long int i = 180000000L;

    printf("____", a); /* Print the value of a */
    printf("____", s); /* Print the content of s */
    printf("____", p); /* Print the value pointed to by p */
    printf("____", c); /* Print the value of c */
    printf("____", p); /* Print the value of p */
    printf("____", d); /* Print the value of d */
    printf("____", i); /* Print the value of i */

    return 0;
} /* end main */
```

c. State which of the following C identifier names are valid or invalid. (5 MARKS)

Identifier Name

- i. Return
- ii. _C
- iii. aB
- iv. 2c
- v. Malaysia Kita

3. a. What will be the output of the following program?

(1 MARK)

```
#include <stdio.h>
int main()
{
    char val=1;
    if(val==0)
        printf("JAVA");
    else
        printf("PYTHON");
}
```

- b. The following code contains errors. Correct the errors and determine the output.

(8 MARKS)

```
int x = 3, y = 5;
if (x > y);
    printf("A");
if (x = 4)
    printf("%d\n", x+y);
```

- c. What will the following program print when it runs? Identify the program's output at each step.

(8 MARKS)

```
int main(void)
{
    int main(void) {
        int a = 5;
        char s[] = "Classical Music";
        char* p = (s + 4);
        char c = 'X';
        double d = 150000;
        long int i = 180000000L;

        printf("____", a); /* Print the value of a */
        printf("____", s); /* Print the content of s */
        printf("____", p); /* Print the value pointed to by p
*/
        printf("____", c); /* Print the value of c */
        printf("____", p); /* Print the value of p */
        printf("____", d); /* Print the value of d */
    }
}
```

```

printf("____", i); /* Print the value of i */

return 0;
} /* end main */

```

- d. What will the following program print when it runs? Show how you computed the results of each operation. (6 MARKS)

```

int main(void)
{
    int x=17;
    int y=24;
    int p=(x & y);
    int r=(x | y);
    int s=(x ^ y);
    int t=(y >> 2);
    int v=(x << 3);
    int a= 42;

    printf( "%d\n", p );
    printf( "%d\n", r );
    printf( "%d\n", s );
    printf( "%d\n", t );
    printf( "%d\n", v );
    printf( "%d\n", a );
    return 0;
} /* end main */

```

- e. What will be the output of the following program?

(2 MARKS)

```

#include <stdio.h>
#define scanf "%s Hide and Seeks"
main()
{
    printf(scanf, scanf);
    getchar();
    return 0;
}

```

4. a. What is a function in C programming? (3 MARKS)
- b. Discuss **FOUR (4)** types of function in C. (8 MARKS)
- c. Write a C program that adds two integers using a user-defined function. (5 MARKS)
- d. Write a program in C to find the sum of all elements of the array. (9 MARKS)

Output:

Input the number of elements to be stored in the array:2

Input 2 elements in the array:

element - 0 : 2

element - 1 : 5

The sum of all elements stored in the array is : 7

-END OF QUESTION PAPER-

```

        printf(" ____", i); /* Print the value of i */

        return 0;
    } /* end main */

```

- d. What will the following program print when it runs? Show how you computed the results of each operation. **(6 MARKS)**

```

int main(void)
{
    int x=17;
    int y=24;
    int p=(x & y);
    int r=(x | y);
    int s=(x ^ y);
    int t=(y >> 2);
    int v=(x << 3);
    int a= 42;

    printf( "%d\n", p );
    printf( "%d\n", r );
    printf( "%d\n", s );
    printf( "%d\n", t );
    printf( "%d\n", v );
    printf( "%d\n", a );
    return 0;
} /* end main */

```

- e. What will be the output of the following program? **(2 MARKS)**

```

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#define scanf "%s Hide and Seeks"
main()
{
    printf(scanf, scanf);
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}

```

4. a. What is a function in C programming? **(3 MARKS)**
- b. Discuss **FOUR (4)** types of function in C. **(8 MARKS)**
- c. Write a C program that adds two integers using a user-defined function. **(5 MARKS)**
- d. Write a program in C to find the sum of all elements of the array. **(9 MARKS)**

Output:

Input the number of elements to be stored in the array:2

Input 2 elements in the array:

element - 0 : 2

element - 1 : 5

The sum of all elements stored in the array is : 7

-END OF QUESTION PAPER-